VICTORY 2026

GEOGRAPHY

Waste Management



Lecture 01

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Topics *to be covered*

1. Need for Waste Management



12. Waste Management

- *Impact of waste accumulation - spoilage of landscape, pollution, health hazards, effect on terrestrial, aquatic (fresh water and marine) life.*
- *Need for waste management.*
- *Methods of safe disposal - segregation, dumping and composting.*
- *Need and methods for reducing, reusing and recycling waste.*

WASTE –TYPES OF WASTE

Waste refers to any discarded material which no longer remains useful. According to the Environment Protection Act, 1990, waste is any substance which constitutes a scrap material, or an effluent or an unwanted surplus which arises out of the application of any process.
There are three types of wastes—solid, liquid and gaseous wastes.

Solid Wastes: It include garbage, food leftovers, decaying fruits and vegetables, cans, bottles and ashes.

Liquid Wastes: It refers to sewage discharged from houses, hospitals, restaurants, offices and factories. Oil spill is also a liquid waste.

Gaseous Wastes: It includes fuel exhausts containing carbon dioxide, carbon monoxide, sulphur dioxide etc. Smog is an example of gaseous waste.

TYPES OF WASTE: COMPREHENSIVE CLASSIFICATION & EXAMPLES



AGRICULTURAL WASTE

- **FARMING & LIVESTOCK WASTE**
  Organic materials from agricultural activities.
- **Examples:**
  - Crop Residues (Stalks, husks)
  - Animal Manure & Slurry
  - Pesticide Containers
  - Plastics (Mulch films)
  - Dead Animals
  - Harvest Waste



CONSTRUCTION & DEMOLITION (C&D) WASTE

- **BUILDING & INFRASTRUCTURE DEBRIS**
  Generated from eonstruction, renovation, and demolltion sites.
- **Examples:**
  - Concrete & Bricks
  - Wood & Timber
  - Metals (Steel, copper)
  - Plastics (Pipes, insulation)
  - Glass (Windows)
  - Asphalt
  - Gypsum Board



BIOMEDICAL WASTE

- **MEDICAL & HEALTHCARE WASTE**
  Infectious, sharps, human/animal tissue.
- **Examples:**
  - Used Syringes & Needles
  - Soiled Bandages & Dressings
  - Pathological Waste
  - Expired Medicines
  - Surgical Gloves
  - Blood & Body Fluids



RADIOACTIVE WASTE

- **NUCLEAR & MEDICAL SOURCES**
  Emits radiation; requires specialized handling.
- **Examples:**
  - Spent Nuclear Fuel
  - Low-level Waste (Lab materials)
  - High-level Waste (Reactor components)
  - Medical Isotopes Waste
  - Uranium Mining Waste



MUNICIPAL SOLID WASTE (MSW)

- **DOMESTIC & COMMERCIAL WASTE**
  Household trash, office papers, packaging, food scraps, textiles, yard waste.
- **Examples:**
  - Food Waste (Peels, leftovers)
  - Paper & Cardboard
  - Plastics (Bottles, bags)
  - Glass
  - Metals (Cans)
  - Textiles (Old clothes)
  - Garden Waste (Leaves, branches)

GLOBAL WASTE STREAM

IINDUNDRIAL WASTE



INDUSTRIAL WASTE

- **MANUFACTURING & MINING PROCESSES**
  Scrap materials, sludge, slag, chemicals, debris from industries.
- **Examples:**
  - Chemical Residues
  - Slag & Sludge
  - Scrap Metals
  - Industrial Solvents
  - Mining Tailings
  - Ash
  - Packaging Materials



HAZARDOUS WASTE

- **TOXIC, FLAMMABLE, CORROSIVE, REACTIVE**
  Threat to human health & environment.
- **Examples:**
  - Pesticides & Herbicides
  - Paints & Solvents
  - Batteries (Lead-acid, lithium)
  - Medical Waste (Infectious)
  - Industrial Chemicals
  - Mercury-containing items



ELECTRONIC WASTE (E-WASTE)

- **DISCARDED ELECTRICAL DEVICES**
  Contains valuable and hazardous materials.
- **Examples:**
  - Mobile Phones & Tablets
  - Computers & Laptops
  - Televisions & Monitors
  - Printers & Scanners
  - Batteries & Chargers
  - Circuit Boards



PROPER MANAGEMENT IS CRUCIAL FOR ENVIRONMENTAL SUSTAINABILITY

Ques

Used computers, tablets, printers are example of ——

(a) Nuclear Waste

(b) Medical Waste

(c) Electronic Waste

(d) Domestic Waste

IMPACT OF WASTE ACCUMULATION

SPOILAGE OF LANDSCAPE

Spoilage of landscape refers to the accumulation of of garbage and waste that is simply dumped onto vacant land and left to decompose, which not only ruins the natural beauty of the land but also provides a home to rats and other disease carrying organisms. Both open dumps and landfills may contain poisonous substances that move into the groundwater or flow into streams and lakes. These open dumps give bad smell and also spoil the beauty of a place .

POLLUTION

The word 'pollute' means to degrade or to contaminate. Pollution is thus, an unfavorable modification of the natural world caused entirely

1. Pollution is caused by human activities like addition of toxic chemicals through the atmosphere into the biosphere.

2. Accumulation of waste is the most visible form of pollution. Every year human beings dispose of billions of metric tonnes of solid waste.

3. The handling of solid wastes is a problem because most disposal methods cause harm to the environment. Both open dumps and landfills may contain toxins that seep into the soil and the water bodies and cause soil and water pollution.

Sources of Pollution			
Air Pollution	**Water Pollution**	**Soil Pollution**	**Radioactive Pollution**
Automobiles	Domestic Effluents	Industrial Wastes	Nuclear Weapons
Chemical Industry	Sewage	Commercial Wastes	Atomic Reactors and Nuclear Fuel
Metallurgy	Industrial Effluents	Domestic Wastes	Radioactive Isotopes
Pesticides	Agricultural Effluents	Chemical Fertilizers	Radiations from X-rays
Refineries	Radioactive Wastes	Biomedical Waste	
Forest Fires	Thermal Pollution	Pesticides	
Burning of Garbage	Offshore Drilling and Oil Spills	Insecticides	
Energy Production Construction Material			

1. Uncontrolled burning of waste creates smoke and other air pollutants -cause air pollution.

2. Scavengers and stray animals invade the open garbage dumps and spread the waste over a large area, thereby, spreading germs and diseases as well as destroying the beauty of the place.

3. Industrial waste contains harmful chemicals, particulate matter (small particles) and toxic heavy metals such as lead and mercury. These toxic chemicals and heavy metals get deposited in animal tissues and harm living things along the food chain. For example, grass gets some toxic chemicals from the soil. Animals eat such grass and get affected by toxins.

4. These animals directly or indirectly pass on these toxins to the human beings through their dairy products or meat.
5. As accumulated waste decomposes, it produces a large quantity of methane gas. This is highly inflammable and can cause an explosion if not managed properly.

WATER POLLUTION

1. Water pollution occurs when people discharge large amount of waste into water bodies.





Eutrophication

1. **Eutrophication** is the process of depletion of oxygen from water bodies occurring either naturally or due to human activities.

2. The process of eutrophication takes place due to introduction of nutrients and chemicals through discharge of domestic sewage, industrial effluents and fertilizers from agricultural fields.

3. Under normal conditions, algae and phytoplankton use carbon dioxide, inorganic nitrogen and phosphate from the water as food. They serve as food for microscopic animals (zooplankton). Small fish feed on these zooplanktons and large fish in turn consume these small fish. When nutrients become abundant due to waste accumulation, the growth of phytoplankton and algae increases.

4. They reduce the penetration of oxygen, light and heat into the water body. As a result aquatic plants are unable to carry on photosynthesis reducing the oxygen content of water. This causes death of most of the aquatic organisms, draining water of all its oxygen.

HEALTH HAZARDS

Spread of Disease Through Contamination

1. Waste that is not properly managed is a serious health hazard.

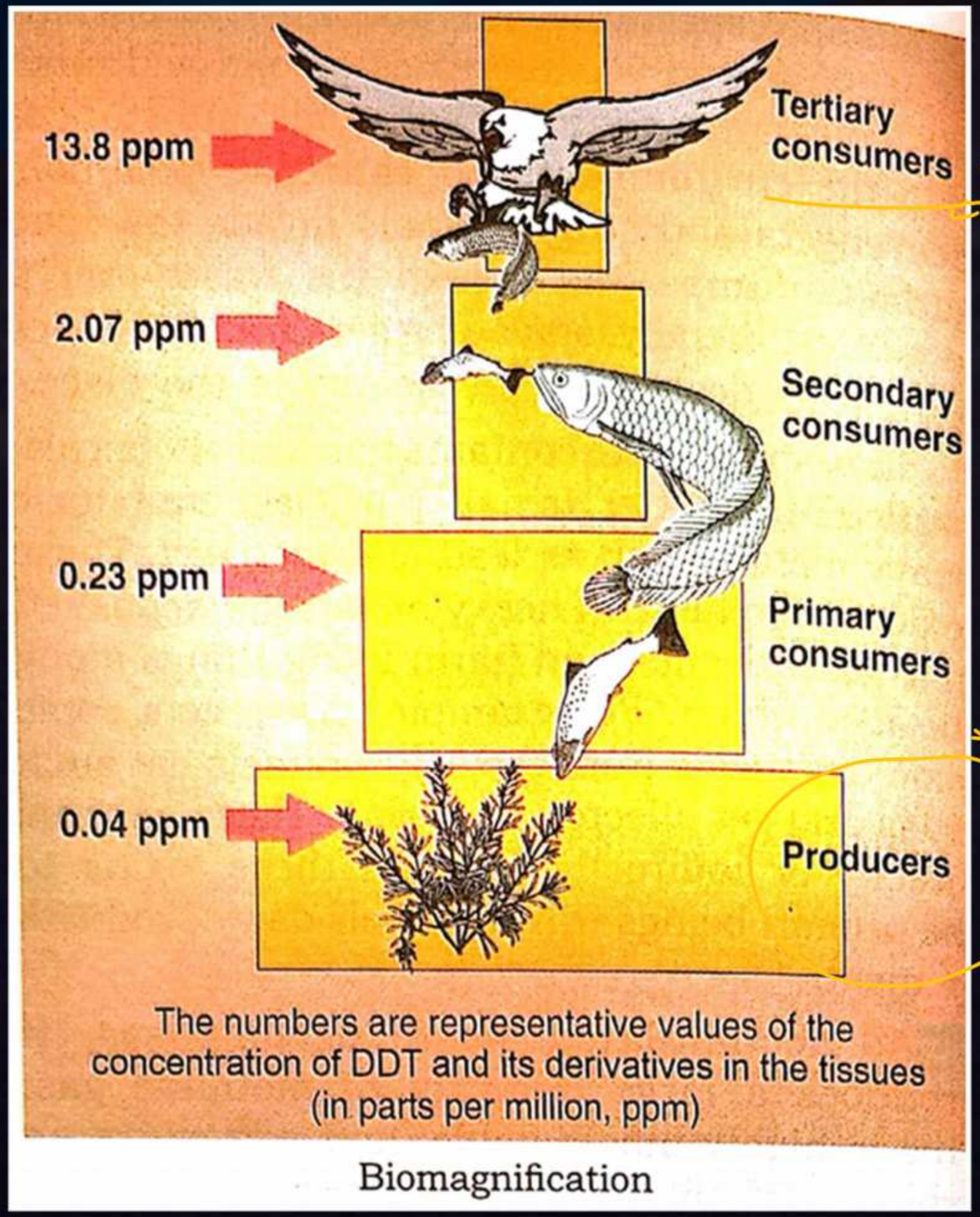
2. Unattended waste dumped in the open attracts flies, rats and other creatures that act as vectors of the diseases and spread them among human beings.

3. Domestic waste poses a serious threat since it is organic in nature, it undergoes fermentation and creates conditions favourable for the survival and growth of pathogens. (disease-causing).

4. Flies, insects, rodents, etc., live in the accumulated waste heaps and carry germs of various diseases to human habitations.

5. Decomposition of wastes produces harmful gases that pollute the air around us.

- Waste dumped near a water source percolates through the soil into the water bodies and contaminates the water.
- Direct dumping of untreated waste in rivers, seas and lakes results in the accumulation of toxic substances in the water bodies and further in the food chain through plants and animals that feed on it (biomagnification).
- Choking of drains and gully pits by the solid wastes results in water logging, especially during the rainy season.
- The water logging results in breeding of mosquitoes in the stagnant water which spread diseases like malaria and chikungunya.



Biomagnification

1. Hazardous wastes are toxic substances which cause an increase in death rate and serious illness.

2. Lead is a harmful toxin and it can affect the development of a child's brain. Asbestos can cause a respiratory disease known as asbestosis, as well as chest and lung cancer.

3. Mercury is a highly toxic chemical which attacks the nervous system, causing brain damage and even death. Arsenic is another chemical that has been shown to cause cancer.

4. Radioactive waste produced by nuclear reactors and weapon factories causes a potentially serious environmental problem. Radioactive waste, although present in small quantities, remains extremely harmful to human health for many years.



Effects of Radioactive Waste

EFFECT ON TERRESTRIAL LIFE

Terrestrial life includes all the organisms that live on land – human beings, plants and animals.

- **Effect on Human Beings:** Accumulation of solid waste looks ugly, smells foul, attracts insects, rats and other animals that spread diseases. Burning of waste in the open dumpyards causes smoke and bad smelling air. In addition, rainwater can drain through refuse and carry harmful substances to different places.
- Sanitary landfills are not fit for human settlements because methane and carbon dioxide gases are released in the first two years. These gases are produced when solid wastes start decomposing underground.
- **Effect on Plants:** Waste accumulation has dangerous effect on plant life. Plant life is affected either by direct deposition of harmful toxins from wastes or indirectly through soil. The toxins cause:

- The toxins cause:

(i) different types of leaf injuries.
(ii) premature leaf fall.
(iii) decrease in transpiration.
(iv) reduction in the rate of photosynthesis.
(v) reduction in biological nitrogen fixation.
(vi) dust deposited on leaves block the stomata of plants. These decrease the rate of transpiration and inhibit the absorption of nutrients from soil.
(vii) smoke emitted by burning of waste causes reduction in root and shoot lengths, number of leaves and number of grains per spike in case of crops like wheat.

Effect on Animals and Birds:

1. Scavengers and stray animals like dogs, rats, pigs and cows are directly affected by waste when they feed on the waste for food.

2. Sometimes these animals consume toxins or non-degradable substances like plastic carry bags present in the waste and die due to choking.

3. The wastes consumed by animals also lead to many diseases and other problems.



- Birds feeding on agricultural waste have thinner and weaker egg shells and increased mortality.



EFFECT ON AQUATIC LIFE

1. Waste accumulation can cause significant damage to aquatic life, both fresh water and marine.

2. Two categories of waste that cause greatest damage to aquatic life are - **pesticides, which run off agricultural lands and industrial and domestic wastes that are improperly disposed of into water bodies**.

3. Over one million marine animals including mammals, turtles, fish, sharks, etc. are killed each year due to plastic waste in the oceans.

4. The plastic waste is washed out from our beaches, streets and highways into rivers and streams and finally into the oceans.

5. After some time, this plastic waste breaks down into smaller pieces, which are easily ingested by marine animals causing blockages in their digestive tracts and eventually death. Marine turtles are more prone to the effects of consuming marine waste

1. Drilling for oil also causes harm to the marine animals in the form of pollution from oil spills.



2. Toxins present in the wastes can kill aquatic organisms directly by sufficiently changing the pH of water, covering the water surface and causing a reduction in dissolved oxygen. **The aquatic organisms are also affected by a phenomenon, known as biomagnification.**

3. **Biomagnification refers to the increase in the concentration of various toxic substances along the food chain.**

4. Toxic substances at the level of primary producers get concentrated at each trophic level as they move up the food chain. The phenomenon of concentrated toxic deposition at the higher trophic level is known as bio-accumulation.

1. A small amount of toxic constituent which is neither excreted nor metabolised, gets increased as the food chain moves upward from one trophic level to the next and the toxic constituents become concentrated.

2. For example, if there are traces of toxic chemicals in water, then their concentration in algae will be much higher. When fish eat the algae, the concentration of toxins will increase further.

3. It will increase still further when big fish will eat smaller fish. Therefore, accumulation of a small amount of toxic chemicals in water can have a serious impact on the fish that live in it.

1. Carelessness and the deliberate dumping of wastes and oil spills in the seas and oceans pollute water and damage beaches.

2. Marine pollution is a great threat to sea-life (plants and animals). Oil spills decrease the penetration of light and hamper the photosynthesis process.

3. They also retard the rate of oxygen uptake by water and adversely affect the development of marine organisms, increase their susceptibility to disease and affect their reproductive processes.

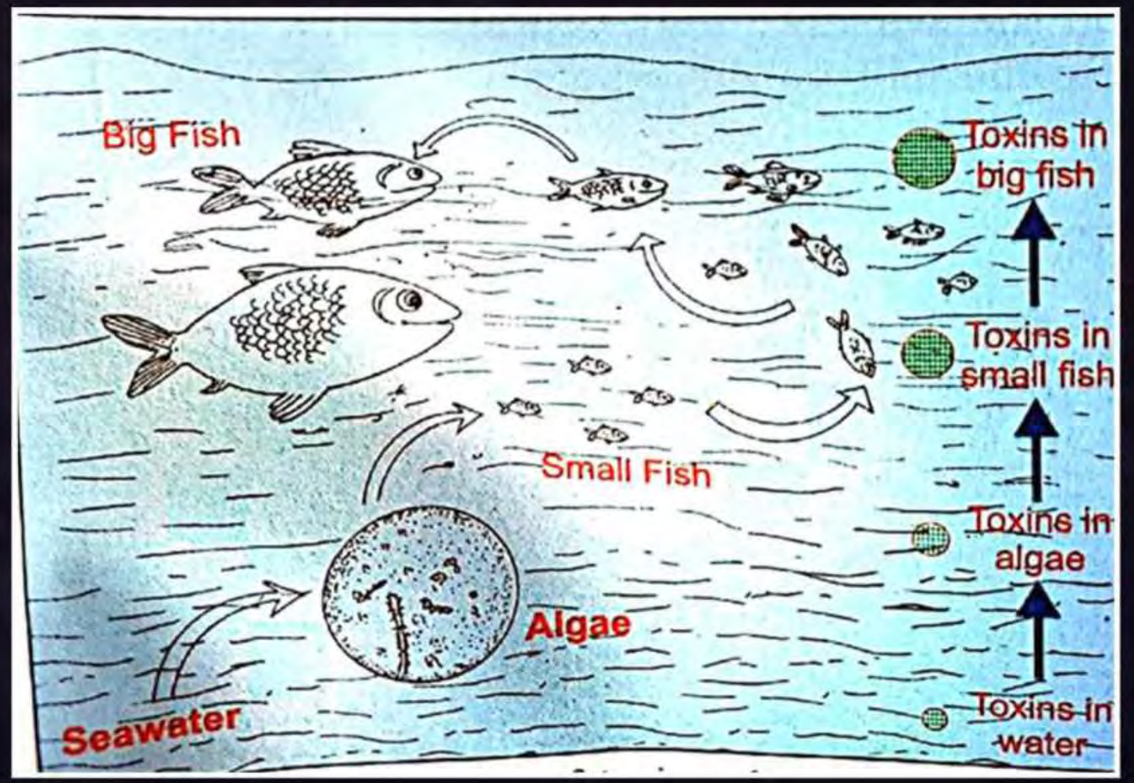
4. They also lead to gastrointestinal irritation, liver and kidney problems and damage the nervous system of marine mammals.

1. Mercury in water can affect marine animals adversely.

2. Mercury contamination in sea on a coastal town in Japan caused illness in some fishermen because of consuming fishes caught from the Minamata Bay.

3. It also killed different sea bird species feeding on fishes. This epidemic in Minamata is known as Minamata Disease.

4. Mercury contamination also results from wastes of other industries like paper and pulp industry, chlorine industry, pesticide industry etc. Pesticides with mercury add a considerable proportion of mercury to natural water bodies.



NEED FOR WASTE MANAGEMENT

- Waste remains in our environment. We move waste from one place to another but never get rid of it completely. Nature has recycled waste materials for millions of years.
- However, human population has increased so rapidly during the last century that the environment is now threatened by our activities and the wastes they produce.
- Much damage is done to the environment by the pollution of air, degradation of soils and contamination of water sources such as rivers, lakes, etc.



- Many scientists believe that if we do not learn to use our resources carefully and to reduce waste, we will not be able to survive.
- There is a close connection between waste, pollution and the damage to the environment.
- Waste causes pollution, which in turn causes the damage. Hence, there is an urgent need for management of waste.

TRANSMISSION OF DISEASES

WASTE ON LAND: Vectors like flies, mosquitoes, rodents and pet animals transmit these diseases. The waste is a breeding ground for such vectors. Hence, there is an urgent need to manage waste disposal in an effective manner.

1. Housefly: Typhoid, diarrhoea, dysentery, cholera, gastro-enteritis, etc.
2. Sandfly: Kala-azar, sandfly fever, etc.
3. Tsetse fly: Sleeping sickness.
4. Mosquitoes: Malaria, filaria, yellow fever, dengue, chikungunya, encephalitis, etc.
5. Rodents: Plague, salmonellosis, etc.
6. Pet animals:

(a) Dog -Rabies, hydatid disease(parasitic infection) etc.
(b) Cat - Dermatophytosis, anthrax, etc.

Ques

_____ disease affected a town of Japan due to Mercury Contamination

(a) Minamata disease
(b) Kala azar
(c) Dengue
(d) Malaria

WASTE IN WATER:

Read

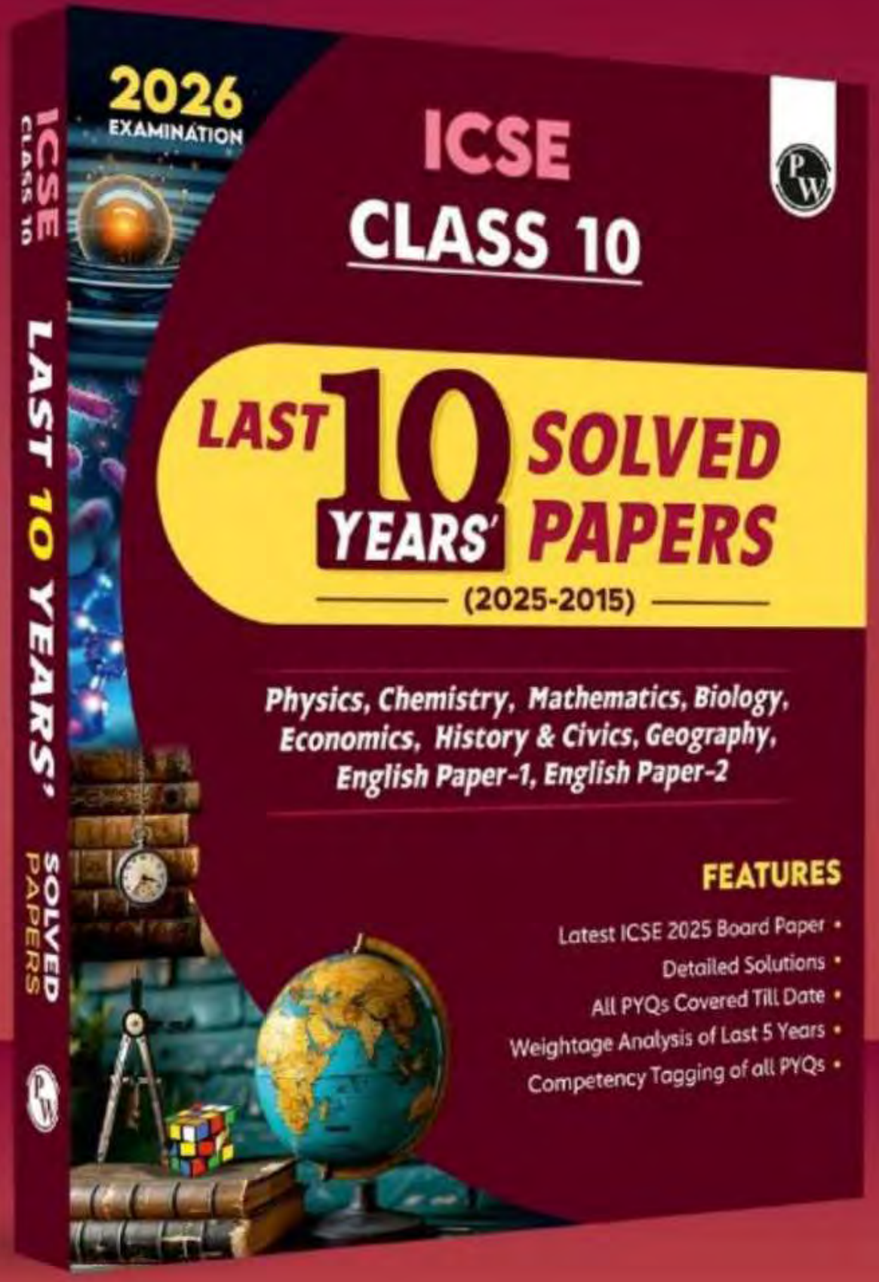
Industrialisation and urbanisation pollute water in the following manner.

(i) Sewage contains organic matter that cannot be decomposed. Also sewage has pathogenic agents.
(ii) Industrial and commercial waste has toxic agents including metal salts and complex synthetic organic chemicals.
(iii) Fertilisers and pesticides produce pollutants.
(iv) There are also other pollutants like radioactive substances.

Human beings are affected by pollution:

(a) by drinking contaminated water,
(b) by using contaminated water for purposes of personal hygiene and recreation.

PHYSICS WALLAH



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